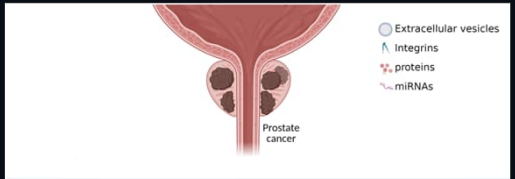
SILAC-TMT-RTS-SPS-MS3 Pipeline

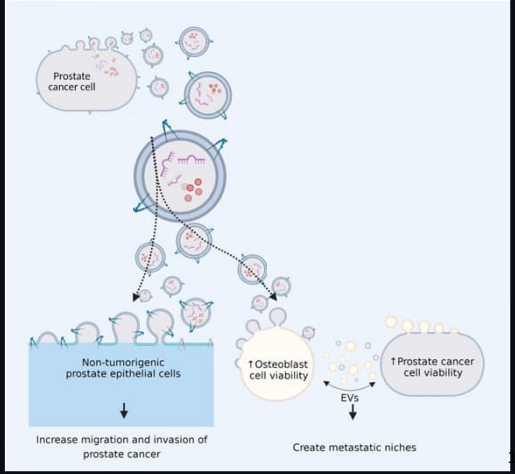
What Drives EV Uptake in the Bone Niche?

Quantitative proteomic profiling of metastatic vs. non-metastatic prostate cancer EVs

```
                    Model:

PC3 (non-metastatic) EVs
PC3M (metastatic)    EVs     → SAOS-2 osteoblasts
```





https://www.frontiersin.org/files/Articles/746040/xml-images/fonc-11-746040-g001.webp

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October 28, 2026

Experimental Setup: Tracing EV Uptake in Bone Niche

Context :

- Prostate cancer preferentially metastasizes to bones.
- Extracellular vesicles (*EVs*) act as messengers to prime this niche.

Core Question:

- Is EV uptake a stochastic, passive event, or are certain vesicle populations preferentially internalized?

Global Aim:

- Understand what cargo features make metastatic EVs more prone to internalization by osteoblasts.

Objective:

- Identify specific cargo proteins differentially enriched in recipient cells to pinpoint molecular candidates driving or co-internalized during EV uptake.

The Cell System:

- Recipient cells: Osteoblasts (*SAOS-2*, mimicking the bone niche).
- EV donors: Non-metastatic (*PC3*) vs. highly metastatic (*PC3M*).

Function	Channels	Sample / Content
Uptake (Non-Met)	130N, 130C, 131N	SAOS-2 + PC3 EV
Uptake (Metastatic)	131C, 132N, 132C, 135N	SAOS-2 + PC3M EV
Spike-in Reference	133N, 133C	SAOS-2 lysate + PC3 EV
	134N, 134C	SAOS-2 lysate + PC3M EV
Negative Controls	128C, 129N, 129C	SAOS-2 alone (background)
	127C, 128N	Empty
Boost Channels*	126, 127N	Pure concentrated EVs

**Boost channels used strictly for MS2/MS3 precursor triggering; excluded from quantitative uptake calculations.*

The Quantitative Approach (SILAC + TMT):

- Donor EVs labeled with Heavy Arg10/Lys-D8 to trace internalized cargo.
- TMTpro 18-plex multiplexing is used to quantify all conditions in a single run.

From Identification to Quantification: Filtering Heavy Cargo

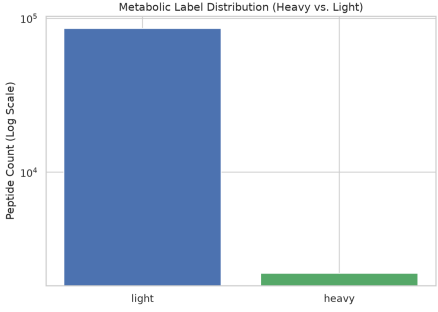
MS1/MS2 Identification Stage

Identification Confidence:

- PSM filtering at $q \leq 0.01$ (FDR 1%).
- All contaminants excluded.

Label Disbalance:

- Endogenous recipient cells (Light): 97.5% (n = 85,929).
- Vesicular cargo (Heavy): 2.5% (n = 2,209).



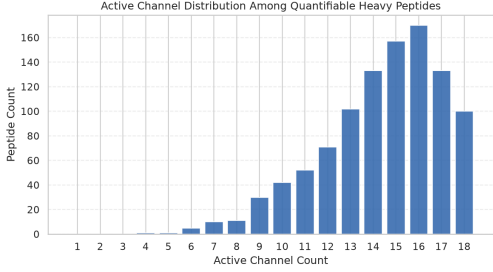
RTS & MS3 Quantification Stage

RTS Duty-Cycle Threshold:

- MS3 triggering targeted strictly heavy precursors.
- 53.2% of heavy peptides reached MS2, but real-time search (RTS) duty cycle prevented MS3 acquisition.
- 0.7% lost to co-isolation interference (low quality).

Yield:

- Retained 1,018 quantified heavy peptides with multi-channel occupancy *(7 of them have signal only in empty and/or spike channels, and not in uptake channels, thus removed)*



Protein Rollup & Noise Gating: Cleaning Technical Artifacts

Technical Outlier Removal

Identification of Preparation Failure (PC3_EV_R1):

- drop in >3.5 times in cumulative reporter intensity (2,300 vs ~8,000 in parallel replicates).
- Median intensity collapsed to instrument noise level (3 vs ~12 in parallel replicates) with 33% zero-value dropouts.
- Thus Excluded from quantitative comparisons to prevent systematic dilution bias:

Channel	Condition	Total Intensity	Median Intensity	Missing Counts
PC3_EV_R1	SAOS2 + PC3 EV	2267	3	123
PC3_EV_R2	SAOS2 + PC3 EV	7448	9	26
PC3_EV_R3	SAOS2 + PC3 EV	8614	11	21
PC3M_EV_R1	SAOS2 + PC3M EV	7074	11	29
PC3M_EV_R2	SAOS2 + PC3M EV	9511	13	11
PC3M_EV_R3	SAOS2 + PC3M EV	10845	15	15
PC3M_EV_R4	SAOS2 + PC3M EV	7100	11	29

Data Curation & Noise Filtering

Protein Rollup & Isoform Specificity:

- 73 heavy peptides mapped to multiple related protein isoforms (e.g., ACTB, TUBA4A, VIM).
- Analysis restricted to unique peptides to eliminate ambiguity -> initial cohort of 372 proteins.
- (Resolving the 20 multi-mapping proteins via gene-level collapsing remains to be done...)

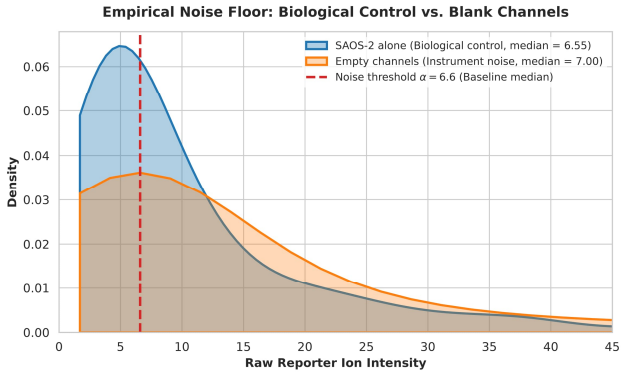
Two Step Background Filtering:

- Absolute Noise Floor ($\alpha = 6.6$): Derived empirically from empty channels and negative SAOS-2 controls (retains 281 proteins).
- Signal-to-Background Ratio ($\beta = 2.0$): Requires $\geq 2 \times$ over SAOS-2 background (3σ confidence based on technical $CV_{ratio} \approx 34\%$).
- Outcome: Yields a high-confidence cohort of 212 authenticated EV cargo proteins.

1. Absolute Noise Floor Threshold (**α**):

The baseline noise profile was empirically estimated using non-zero reporter intensities across negative biological controls (recipient cells (SAOS-2), n = 3) and empty instrument channels (n = 2). Across these background channels, the median signal was found to be 6.55 in SAOS-2 controls and 7.00 in empty channels (with a 75th percentile of 12.90 and 14.90, respectively). Given the near-identical intensity distributions between empty and recipient control channels, this signal represents non-specific baseline noise. Accordingly, the absolute threshold was set to the empirical background median (α= 6.6). A protein was required to exceed this noise floor in at least one experimental condition:

$$\max(Median_{PC3}, Median_{PC3M}) > \alpha$$



2. **Signal-to-Background Ratio (β):**

To account for technical variability and avoid false-positive assignments due to analytical fluctuations, the required fold-change threshold (β) was derived from experimental reproducibility. The median coefficient of variation ($CV = \mathrm{SD}/\mathrm{Mean}$) across the replicates of the recipient control channels was determined to be 24%. Under the standard law of error propagation, the relative technical uncertainty of the ratio between two independent noisy measurements (sample vs. control) is:

$$CV_{\text{ratio}} \approx \sqrt{CV_{\text{sample}}^2 + CV_{\text{control}}^2} = \sqrt{0.24^2 + 0.24^2} \approx 0.34\ (34\%)$$

To achieve statistical confidence at the 3σ level ($p < 0.003$), an authentic cargo signal must exceed random analytical fluctuations by at least three standard deviations of the ratio ($3 \times 34\% \approx 102\%$, representing a >2-fold increase). Thus, the fold-change threshold was established at $\beta = 2.0$ ($\log_2 FC \geq 1.0$):

$$\max(\text{Median}_{\text{PC3}}, \text{Median}_{\text{PC3M}}) \geq \beta \times \text{Median}_{\text{SAOS2 control}}$$

From an initial pool of 372 strictly quantified proteins, 281 proteins (75.5%) passed the absolute baseline noise threshold ($\alpha = 6.6$), successfully filtering out near-LOD artifacts and low-level instrument fluctuations. Subsequently, applying the dual-criteria filter ($\alpha = 6.6$ and $\beta = 2.0$) retained a cohort of 212 validated EV cargo proteins (55.6% of the initial cohort), while discarding 74 proteins as non-specific background or ambiguous signals within the experimental noise margin.

Spike-in Normalization & Invariance to Particle Dosing

Decoupling Biological Uptake from MS Responsiveness

Problem:

- Raw reporter intensity does not reflect uptake efficiency.

Internal Spike-in Ratio:

$$Relative\ Uptake_{ij} = \frac{I_{Sample,ij}}{mean_j(I_{Spike\ in,ij})}$$

(for protein i in channel j)

Transformation to $\boldsymbol{log_2}$ space :

$$y_{ij} = log_2(Relative\ Uptake_{ij})$$

(Conservative LOD/2 substitution for rare 0.15% dropouts)

Why?

- Anchored to reference standard: recipient cell lysate + pure donor EVs.
- Shifts values from absolute counts to a dimensionless metric.
- Enables direct comparison of uptake affinity ("penetrating capacity") between different protein species.

Resolving Global Shifts & Dosing Uncertainty

Initial Adjustment (NTA Offset):

- NTA showed ~ 2 × more PC3 EVs than PC3M in suspension.
- A nominal scaling offset ($\Delta \log_2 = -1.0$) was empirically applied to account for particle count differences.

The Emerging Global Shift:

- Empirical particle counts carry high heuristic variance.
- Exploratory PCA revealed a pervasive Global Scale Effect, indicating that cumulative uptake differences far exceeded simple particle dosing.

Solution: Sample-Median Centering:

$$y_{ij}^{centered} = y_{ij} - median_j(y_j)$$

(for protein i in channel j)

Dual effect & Mathematical Invariance:

- This method absorbs bulk uptake effects while simultaneously canceling any uniform additive dosing constant C:

$$(y_{ij} - C) - median_j(y_j - C) = y_{ij} - median_j(y_j)$$

- Thus cargo selectivity is decoupled from both bulk endocytosis and heuristic NTA enumeration noise.

Exploratory PCA: Uncovering the Global Scale Effect

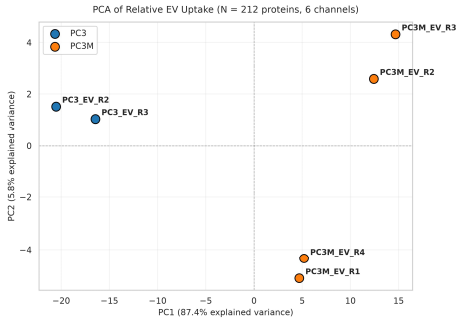
Visualizing Bulk Separation

Phenotypic Segregation:

- Separation between non-metastatic PC3 (n=2) and metastatic PC3M (n=4) replicates is observed.
- High intra-group biological reproducibility (Pearson r = 0.9 for PC3; r = 0.8 – 0.9 for PC3M).

Variance Anomaly:

- PC1 dominates 87.4% of total dataset variance.



Feature Loadings & The Biological Confounder

Unidirectional Bias in Loadings:

- Analysis of PC1 eigenvectors reveals that 211 out of 212 proteins exhibit strictly positive loadings.
- Only a single protein displays a negative loading.

Interpretation:

- In uncentered linear decomposition, a component driven by near-unanimous positive loadings represents a canonical Global Scale Factor.

Implication:

- Metastatic PC3M EVs are internalized by recipient SAOS-2 osteoblasts at a systematically higher absolute volume/rate than PC3 EVs.

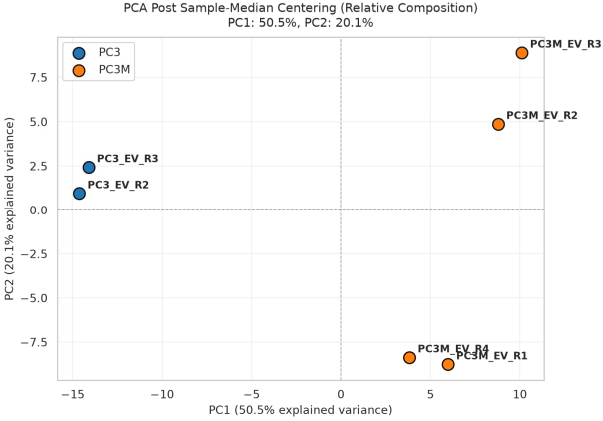
Problem:

- Massive non-specific bulk endocytosis inflates nearly all protein signals, completely masking true stoichiometric cargo selectivity.

Resolving Relative Cargo Selectivity: Post-Centering PCA

Elimination of Global Bulk Confounder:

- Subtracting channel medians standardizes sample baselines to 0.0 in $\log_2$ space.
- Cancels out non-specific internalization volume differences.



Balanced Feature Loadings:

- Unlike the uncentered PCA (where 99.5% were positive in PC1), post-centering loadings reveal a balanced, bidirectional spectrum (99 positive and 113 negative loadings in PC1)

Genes	PC1 Loading	log_2(Fold Change)
SLC12A2	0.097	0.83
CTNND1	0.096	1.97
COL4A2	0.096	6.01
ANPEP	0.096	1.79
TGM2	0.095	4.13
FN1	0.095	3.76
CCN1	0.094	1.71
HLA-A	0.094	1.03
	...	
RPS16	-0.093	-0.87
RPL4	-0.094	-0.82
ATP1A1	-0.095	-0.33
HM13	-0.095	-1.77
TMED9	-0.096	-0.78
PDIA3	-0.096	-0.91
HSPA5	-0.096	-1.04
MAP3K12	-0.096	-2.25

Differential Uptake: Overcoming Small Sample Constraints with Empirical Bayes

Unmoderated Welch's t-Test

Small Sample Size ($\boldsymbol{n_1 = 2, n_2 = 4}$):

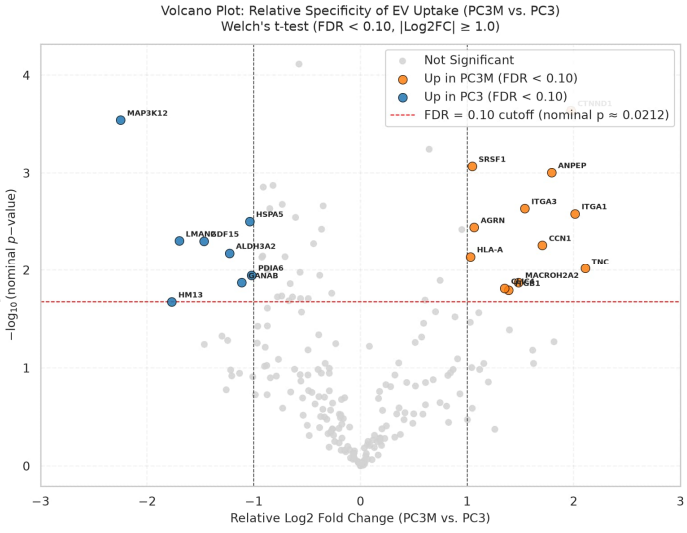
- Residual degrees of freedom for the PC3 arm collapse to $n_1 - 1 = 1$, restricting effective degrees of freedom ν to an unstable range ($1 \leq \nu \leq 3$).

Individual sample variances ($\boldsymbol{s_g^2}$) fluctuate wildly:

- False Positives: Chance agreement between 2 replicates artificially deflates denominator variance.
- False Negatives: Slight sample variance penalizes true large-effect drivers under FDR correction.

Outcome:

- Identified 22 significant proteins



Differential Uptake: Overcoming Small Sample Constraints with Empirical Bayes

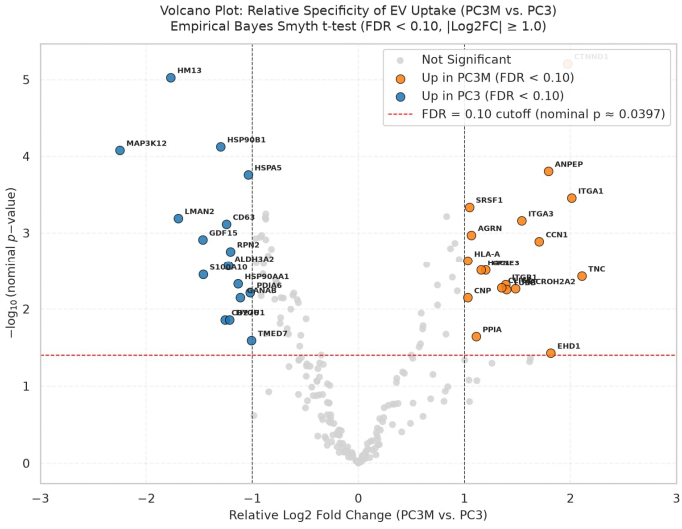
Empirical Bayes Moderation (limma-trend)

Borrowing Information Across the Proteome:

- Prior variance $s^2_{0,g}$ is modeled as a smooth function of mean abundance (A_mean, spline curve).
- Individual variances shrink toward the common trend, augmenting effective degrees of freedom from $d_g = 4$ to $d_g + d_0 \approx 7.1$.

Statistical Recovery:

- Expands the cohort to 37 high-confidence cargo drivers (+68% discovery yield).
- Retains 100% of initial Welch t-test hits



High-Confidence Cargo Signatures: Empirical Bayes Panel (37 Drivers)

Total proteins passing FDR < 0.10 and |Log2FC| >= 1.0: 37							
Up in PC3				Up in PC3M			
Gene	Log2FC	p_value	q_value	Gene	Log2FC	p_value	q_value
HM13	-1.8	0.00001	0.00077	CTNND1	2.0	0.00001	0.00077
HSP90B1	-1.3	0.00008	0.00294	TGM2	4.1	0.00001	0.00077
MAP3K12	-2.2	0.00008	0.00294	COL4A2	6.0	0.00005	0.00266
HSPA5	-1.0	0.00017	0.00461	ANPEP	1.8	0.00016	0.00461
LMAN2	-1.7	0.00065	0.00864	ITGA1	2.0	0.00035	0.00768
CD63	-1.2	0.00077	0.00903	FN1	3.8	0.00036	0.00768
GDF15	-1.5	0.00123	0.01089	SRSF1	1.0	0.00046	0.00864
RPN2	-1.2	0.00176	0.01287	ITGA3	1.5	0.00069	0.00864
ALDH3A2	-1.2	0.00272	0.01599	AGRN	1.1	0.00107	0.01089
S100A10	-1.5	0.00351	0.01862	CCN1	1.7	0.00130	0.01101
HSP90AA1	-1.1	0.00467	0.02198	HLA-A	1.0	0.00233	0.01541
PDIA6	-1.0	0.00609	0.02482	CPNE3	1.2	0.00309	0.01722
GANAB	-1.1	0.00708	0.02597	H4C1	1.2	0.00308	0.01722
HYOU1	-1.2	0.01376	0.04233	TNC	2.1	0.00373	0.01929
CD276	-1.3	0.01378	0.04233	ITGB1	1.4	0.00479	0.02202
TMED7	-1.0	0.02566	0.06885	CLIC4	1.4	0.00526	0.02323
				MACROH2A2	1.5	0.00544	0.02352
				TUBB	1.4	0.00560	0.02374
				CNP	1.0	0.00713	0.02597
				PPIA	1.1	0.02276	0.06185
				EHD1	1.8	0.03729	0.09524

Phenotype Segregation: Unsupervised Hierarchical Clustering

Method & Key Patterns

Clustering Parameters:

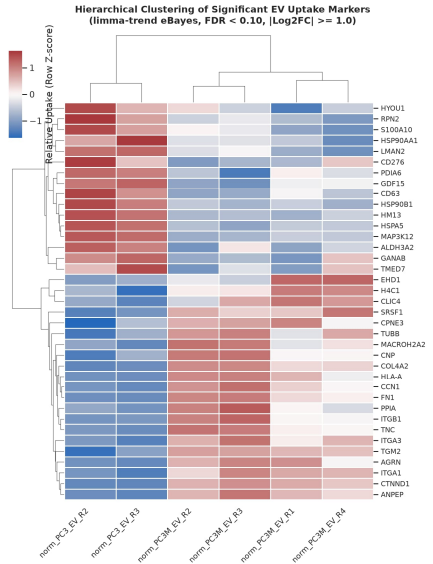
- Metric: Euclidean distance on row Z-scores ($\approx 1 - Pearson\ r$).
- Linkage method: Average (UPGMA).
- Unsupervised: generated without condition labels.

Sample Grouping (Columns):

- Clear split between recipient cells incubated with PC3 (n=2) vs. PC3M (n=4) EVs.
- PC3M replicates cluster into biological sub-pairs (R1/R4 and R2/R3), consistent with PCA.

Observed Protein Clusters (Rows):

- PC3M-enriched cluster (n = 21): Extracellular matrix & adhesion components: COL4A2, FN1, TGM2, TNC, along with integrins (ITGA1, ITGA3, ITGB1).
- PC3-enriched cluster (n = 16): ER chaperones & trafficking markers: HSPA5, HSP90B1, HM13; tetraspanin CD63.



Next Steps

1. Curate Multi-Mapping Peptides:
- Manually group the 73 ambiguous peptides into resolved isoform clusters.
- Recover ~20 discarded proteins without introducing individual assignment bias.

2. Calibrated Pathway Enrichment:
- Run functional enrichment strictly against the experimentally detected proteome background (N=212/372 proteins), avoiding whole-genome bias.

3. Rank-Shift Analysis (Kelly's Suggestion):
- Rank proteins by abundance in Spike-in vs. Uptake channels.
- Identify low-abundance EV cargo showing disproportionately high internalization efficiency.

4. Surface Drivers vs. Delivered Cargo (Grigory's Hypothesis):
- Determine whether enriched proteins actively trigger uptake (e.g., surface integrins) or are just passively delivered cargo (e.g., collagens).
- Correlate EV surface abundance with internalization rates.

5. Experimental Validation:
- Block candidate surface proteins with specific antibodies to see if EV internalization drops.

Thanks for your attention!
All the materials can be found in my GitHub:



https://github.com/Dudkanist/ev-uptake-proteomics